Iridium Project

Early days of cellular device

Communications technology relied on ubiquitous radio towers that provided coverage called "cells". Which meant...

- Towers had x amount of range
- Lack of towers
- Lack of reception in areas with no towers
- Could not build towers with physical obstacles (eg. mountains and tunnels)

The New Era

"Why not flip the whole thing over so it hangs overhead?" _ Bertiger

Put up a satellite to be the cells where all geographical locations can receive reception

History

Iridium was formed in the late 80's by three talented/ambitious engineers by the names: Ray Leopold, Bary Bertiger, and Ken Peterson. It was a sub-company (child Company) of Motorola Space and Systems Technology Group. They planned the first satellite-based mobile communication system. 77 satellites were originally planned to be deployed.

Issue #2
Escalating Commitment

- The Iridium management was aware of the issues plaguing the release of the new system, but continued to press ahead as fear of significant loss motivated their actions.

Tyler Graham

The Motivation

- The Motorola team was heavily influenced by the idea that innovative technology will sell regardless of limitations because of the “neat” factor the product has been associated with.
- While the ability to talk to others world wide seemed like a great idea, it was not as innovative as they thought.
- With the consumer cell network expanding alongside the development of Iridium, the project was losing its flare and most importantly its target market base

Known Issues

- The management insisted on launching the satellite network, the most expensive aspect of the project, even though phone development was not complete and a marketing campaign was barely connected.
- The management is faced with a problem after the costly development and launching of 66 LEO satellites:

They perceive that the costs that went into the Iridium project are too high to end the operation.

Sunk Costs

- The issue that motivated the Iridium lead is referred to as the Sunk Costs Fallacy.
- This is a psychological concept that many can identify with, but most commonly plagues a project manager.
- The Iridium case study exemplifies the concept very well, but let's have a real-world example to relate to the Iridium failure.

In-Depth: Sunk Costs

- The sunk cost fallacy can be applied to almost any point in someone's life where they have **"gone too far too stop now".**
- If I watch a professional baseball game and your favorite team, the team that always loses, magically tied the game into an 11th inning, I know deep down they have no chance of winning.

But I have spent too much time watching the game thus far, so I cannot leave until it is over.

Iridium's Case

- Iridium faced a similar problem, except it was monetary value versus time as in the previous example.
- The management initially sat on billions in investments to fund the research and development.
- At the time the satellite technology had been finished on the ground, too much money had been invested in Iridium to admit the phones were not yet ready and failure was imminent.

Summary

- Iridium management teams were motivated by possible loss and spent much more money and time during Iridium's decline to stop the decline which is also known as "chasing your bets".
- It feels like common sense to simply not spend more money during a failure, cutting the project off currently before significant loss occurs, but human nature indicates otherwise.

Issue #3

Lack of Planning.

Jason Choi

Time Frame

- Project took 11 years to complete.
- Cellular technology spread faster than predicted.
- Targeted only “Business Travelers”.
- “Business travelers” started to use cell phones which had spread the over time. More cost efficient and convenient it terms of the location the device can be used. In short, a better value than Iridium.

Jason Choi

Technology

- Iridium uses Line-of-Sight technology which disabled the subscribers to use the device while in moving cars, inside buildings, and most urban areas.
- Even in Open fields, the end user had to align correctly to get good connection.
- Lacked data capabilities, vital for business travelers.

Jason Choi

QUALCOMM
Globalstar
Looking for
service...
menu
Clr
end
info
send
Pwr
Globalstar

Jason Choi

Design

- Battery recharging in remote areas required special solar-powered accessories.
- Size of brick
- “It’s huge! It will scare people. If we had a campaign that featured our product, we’d lose.”

Jason Choi

Reliance vs Planning

- The belief that innovative technology would eventually attract customers, in fact, was deeply ingrained in Motorola's culture.

Jason Choi

Bottleneck Effect

- Management launched the service before enough phones were available.
- Software problems arose
- Iridium couldn't even get phones to the few subscribers that actually wanted one.
- "(John Richardson) claimed to be vociferous in board meetings, arguing against the November launch. Neither the service, nor the service providers, were ready. Supply difficulties meant that there were few phones available in the market."
- Decided to launch regardless of problems.

Jason Choi

Practicality

- "you can't expect a CEO traveling on business in Bangkok to leave a building, walk outside on a street corner, and pull out a $3,000 phone."

Jason Choi

Summary

- a highly leveraged capital structure
- design limitations -- including phone
- size
- service limitations -- including
- severe degradation in cars, buildings,
- and urban areas
- high handset and service pricing
- the build-out of cellular networks
- a lack of control over partners'
- marketing efforts

Jason Choi

Facts

We need to find out what can be done to make the project a success, and increase profit, and what can be done to prevent the bankruptcy.

Design

- Time to give the designing team a visit. "good job guys on bringing out the phone, but we need to make some modifications to the design. It needs to be smaller.